CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

# Hydroponics Flow Lab

**Goal:** Model a gravity-fed hydroponic path that distributes water evenly without leaking or flooding.

35 minutes | 8 steps | Engineering and Design | Playbook A meeting 23

Ask | Plan | Build | Test | Improve

## You are done when

**Show the measured flow result and explain the one change that made distribution more even.**

## You will use

- Clean tubing
- Cups
- Reclaimed channels
- Water trays
- Measuring cups
- Towels
- Plant-site models
- Teacher-controlled tubing cutter
- Wet mats
- Sink bucket

## Steps

35 minutes total. Your teacher says when to move on.

### 1 Step 1: Get ready — 3 min

- Place the dry tray on the wet mat.
- Measure the test water.

### 2 Step 2: Wet tool safety — 4 min

- Keep water inside the tray.
- Wipe each spill with a towel.
- Do not cut tubing yourself.
- Keep electronics away from the wet zone.

### 3 Step 3: Plan the flow — 3 min

- Draw the water cup.
- Draw the water path.
- Mark each plant spot and the end cup.

### 4 Step 4: Build the dry model — 7 min

- Mark the cut line on the tube.
- Give the tube to the teacher.
- Wait for the cut, then connect the tube path.
- Check every connection while dry.

### 5 Step 5: Run the water — 6 min

- Pour the measured water into the water cup.
- Watch each plant spot.
- Measure the collected water.

### 6 Step 6: Change the flow — 6 min

- Change one slope, hole, or loose connection.
- Repeat with the same water amount.
- Record the new collected amount.

### 7 Step 7: Dry the lab — 4 min

- Drain the tray into the sink bucket.
- Place wet parts on the drying tray.
- Dry the table and floor.

### 8 Step 8: Exit check — 2 min

- Write the collected amount.
- Name the flow change.

## What to turn in

A measured flow result and one change that made the flow more even.

## No computer? Another way to do it

A dry path model with one teacher pour demo.

## Finished early? Next challenge

Flow Alarm: Design a visible marker that shows when one plant site receives too much or too little water.

## Stay safe

Keep water away from outlets and electronics, wipe spills immediately, and use no pump unless the teacher controls it.

## Grade targets

**Grade 5:** Measure water at each plant site with a cup. Mark more or less.
**Grade 6:** Calculate the percent of water recovered and name where it was lost.

## Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.